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SECTOR AT A GLANCE

One Health

Health Sector in India

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One Health

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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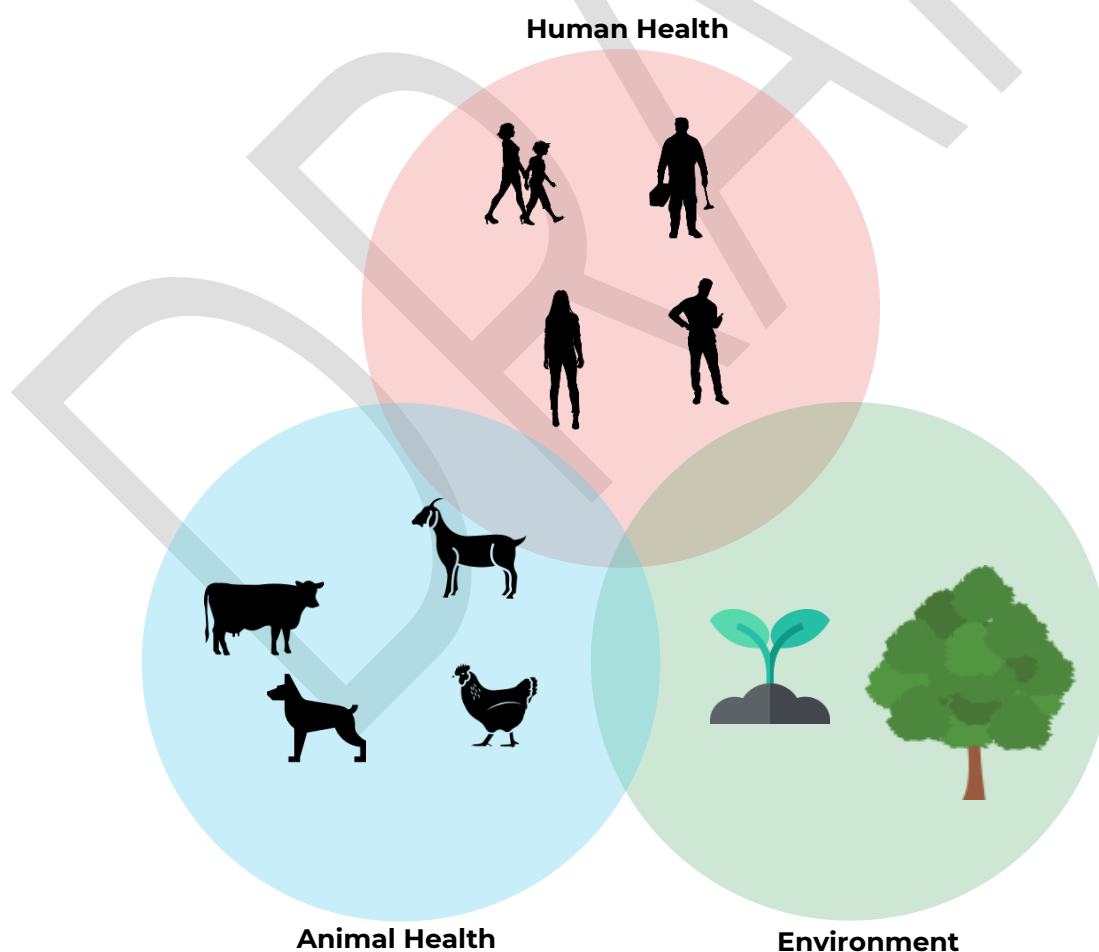
Overview

According to the **World Health Organisation (WHO)** “One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals and ecosystems.” It is an approach that recognises that the health of human, animals and the environment are interconnected. It emphasises on the interdependence of the three domains and seeks to address health issues by considering the complex interactions.¹

Background of One Health Approach²

India first recognised the One Health approach as a response to the Avian Influenza (H5N1) pandemic. In December 2007, India hosted the International Ministerial Conference on Avian and Pandemic Influenza in New Delhi, with participation from 111 countries and support from international organizations such as WHO, FAO, OIE, World Bank, and UNICEF. During this conference, India expressed support for the One Health concept, emphasising an integrated approach to animal and human health. It was acknowledged that promoting One Health could extend the global response to H5N1. India's journey toward One Health began in 2005 when it developed preparedness and contingency plans for H5N1, setting the stage for its One Health efforts.

Figure 1 - One Health³



FAO: Food and Agriculture Organization, OIE: World Organisation for Animal Health, UNICEF: United Nations Children's Fund

¹World Health Organisation. *One Health*. Retrieved 20 September 2023.

² Nambiar P (2020) India to envision One Health movement for confronting emerging health threats: From concept to approach toward institutionalization, *Int. J. One Health*, 6(2): 165-176.

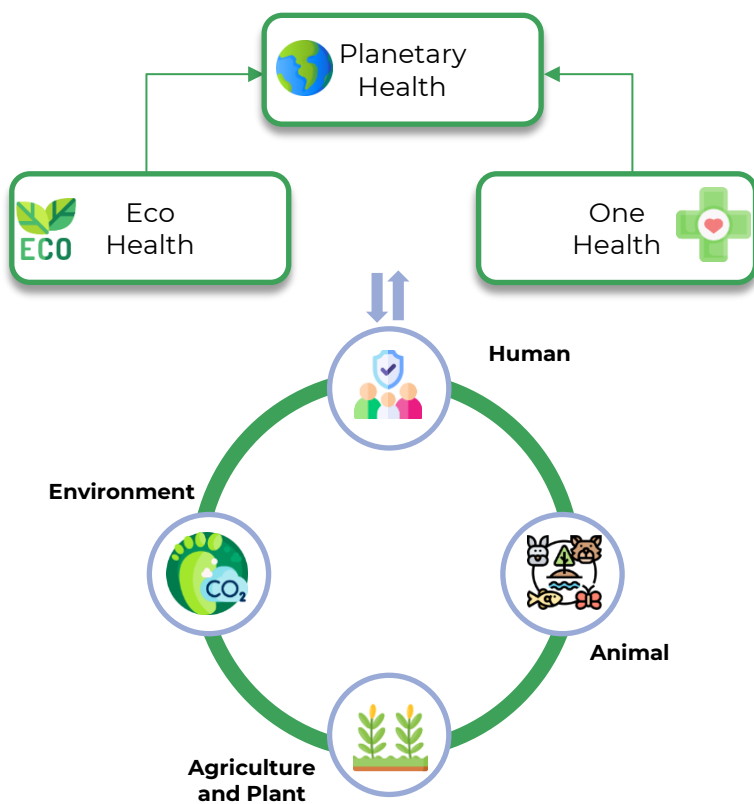
³ Centres for Disease Control and Prevention. (n.d). *One Health Graphics*. Retrieved 20 September 2023

Historical Perspective : One Health Approach

One Health is not a new idea. Greek physician **Hippocrates** stated far back in the times that there is a cross relationship between the airs, waters, and places. Later, **Dr. Rudolf Virchow**, gave a rule in the nineteenth century that there is no separating line between animal and human infection and human medicines, and there should be no such boundary. **Dr. William Osler** has done several studies to establish linkages between the human and animal health. In July 2007, “Father of Veterinary Public Health” **Dr. James H. Steele** further extended the concept of 'One Medicine' to 'One Health, One World'.⁴

Health for All:

When it comes to health for all, the main principles to consider is Planetary Health which includes Eco Health and One Health.



Major Areas for focus on One Health⁶

Pandemic Preparedness

- ✓ Zoonotic Diseases and Vector Borne Diseases
- ✓ Environment Health
- ✓ Antimicrobial Resistance (AMR)
- ✓ Food Safety and Security

Other Areas of Concern

- ✓ Mental Health
- ✓ Chronic Diseases
- ✓ Occupational Health

⁴ OneHealthInitiative. (2022). 'One Medicine-One Health': an Historic Perspective.

⁵ Lerner H. (2017). A Comparison of Three Holistic Approaches to Health: One Health, EcoHealth, and Planetary Health.

⁶ CDC. [n.d]. One Health Basics. Retrieved 31 October 2023.

Evolution of One Health in India²

In India, One Health has evolved through various initiatives and collaborations at national, state, and district levels, involving diverse partners and knowledge systems from health, veterinary, wildlife, agriculture, and other sectors

2005-2007: Genesis of One Health in India



- **Genesis of One Health ideology began in 2007**, however India had sown the seeds for One Health in the year 2005 when its action plan for preparedness, control, and containment of avian influenza H5N1 was developed by the Department of Animal Husbandry, Dairying and Fisheries and the contingency plan for the management of human cases of H5N1 developed by Directorate General of Health Services.²

2009: India Joined with One Health network of South Asia



- **One Health Alliance of South Asia (OHASA)** was launched by the Wildlife Trust to form a cohesive regional network of scientists and policymakers from Ministries of Health, Agriculture and Environment, as well as Nongovernmental Organizations and universities in India, Bangladesh, Pakistan, and Nepal to address zoonotic diseases.²

2010: Launch of National Institute for Zoonoses



- **On July 6, 2010** in Nagpur, experts and scientists from medical and veterinary sciences in a national conference organized by the National Association of Welfare of Animals and Research in collaboration with Indian Medical Association (Nagpur Chapter) conceived the concept of establishing of a National Institute for Zoonoses (NIZ) at Nagpur.⁷



2008 : Growth of One Health Approach

- The World Health Organization, in collaboration with Food and Agriculture Organization (FAO), World Organization for Animal Health (OIE), United Nations Children's Fund (UNICEF), United Nations System Influenza Coordination, and the World Bank developed the framework on 'One Health' in 2008 to decrease the drastic impact of new diseases, and the ones that re-emerged.⁸
- During this time, **Roadmap to Combat Zoonoses in India (RCZI)** initiative was established in June, 2008 in a joint endeavour by the Ministry of Health and Family Welfare, the Ministry of Agriculture and the Wildlife Institute of India, Government of India.⁸



2010 : Centre for Disease Control and Prevention and National Centre for Disease Control Collaboration

- Recognizing the need to broaden cooperation on health, especially emerging global diseases between India and USA, a **Global Disease Detection, India Centre** was established in NCDC.²



2014: Launch of New One Health Centre and One Health course

- Kerala Veterinary and Animal Science University established a Centre for OH Education, Advocacy, Research and Training (COHEART) on February 26, 2014, and also launched two new courses in One Health.²
- Later in 2016, Veterinary Council of India included for One Health concept in Bachelors of Veterinary Science's curriculum for the first time.²

⁷ Chaudhari et al. (2020). Journey towards National Institute of One Health in India.

⁸ Invest India. (2021). One Health: The Evolving Paradigm to Boost India's Healthcare Infrastructure. Retrieved 23 October 2023.

2017: National Action Plan on Antimicrobial resistance incorporates One Health Approach



- National Action Plan on Antimicrobial Resistance (AMR) was drafted in 2016, to address the challenge of rising antimicrobial resistance and efforts were made to incorporate the One Health approach within these plans.²

2019 : One Health India Declaration



- In a conference hosted by the Department of Biotechnology (DBT), Ministry of Science and Technology in partnership with Ministry of Agriculture and Farmers' Welfare and Ministry of Health and Family Welfare decision was taken to launch India's OH Initiative to tackle the most urgent health threats and has also proposed OH roadmap for India.²
- Following that, Delhi declaration was signed by Health Minister of India and other Member States advocating the One Health approach.²

2021-22: One Health Budget and One Health Consortium



- The Union Budget of 2021 specifically allocates a development of National Institution for One Health.⁹
- Department of Biotechnology, Ministry of Science and Technology has launched a One Health Consortium consisting of 27 organisations including medical centres, central government institutes, disease investigation centres, etc.⁹



2019 : Launch of Integrated Health Information Platform

- On November 26, 2018, the Union health secretary unveiled IDSP's (Integrated Disease Surveillance Programme) Integrated Health Information Platform, a platform for tracking public health monitoring.²
- The innovative project gave policymakers access to near-real-time data for disease burden reduction, outbreak detection, morbidity and mortality reduction, and improved health systems.²



2019-20 : Centre of One Health

- In March 2019, the Indian Council of Medical Research (ICMR) declared to establish 'Centre for One Health', a satellite centre under National Institute of Virology, Pune in Nagpur.⁷
- The Department of Biotechnology's National Institute of Animal Biotechnology (NIAB), at Hyderabad, launched the Centre for One Health on November 3, 2020 with the manifesto "Animal health for Human Welfare".⁸



2022 : National One Health Mission and One Health Support Unit

- National One Health Mission was approved on July 20, 2022, with an aim of coordinating, advancing, and integrating all current One Health initiatives across the nation.¹⁰
- To support the Government of India in developing India's National One Health Platform, a One Health Support Unit (OHSU) by Department of Animal Husbandry with Bill and Melinda Gates Foundation.¹¹

⁹ Department of Biotechnology. (n.d). *DBT One Health Consortium*. Retrieved 20 September 2023.

¹⁰ Office of the PSA to the Government of India. (n.d). *One Health*. Retrieved 20 September 2023.

¹¹ One Health. (n.d). *One Health Support Unit*. Retrieved 23 October 2023.

Why India needs to focus on One Health Approach?

India is growing at a fast pace and many factors make India as a hotspot of attracting growing diseases. India has the largest population in the world and accommodates 17% of the world's total population and shares borders with seven countries indicating a higher rate of disease transmission.

India needs to focus on One Health approach to address the risk of zoonotic diseases, safeguard public health, ensure food security, and mitigate economic and environmental impacts. This integrated approach is vital for disease prevention, antimicrobial resistance control, and global health security in an interconnected world.

The areas where One Health Approach is ideal are²:



Zoonotic diseases have gained significant attention with global One Health movement. India being identified as a hotspot with instances of multiple viruses emphasises the importance of One Health.



Antimicrobial Resistance (AMR) is a multifaceted threat to human and animal health, the global economy, and security, necessitating a comprehensive One Health approach. In 2010, India emerged as the country with largest consumption of antibiotics.



A "farm to fork" integrated approach is essential to tackle **food safety challenges** in India, given diverse food habits and factors contributing to foodborne illnesses.



Climate change is adversely impacting the livelihoods of people and animals in India, necessitating interdisciplinary research to understand the **broader effects of environmental health** on human and animal well-being. Collaborative action is essential to address pollution, waste management, and protect marine ecosystems, considering their direct impact on public health and living resources.

The emergence of corona virus pandemic has underscored the significance to reinforce the One Health approach. Neglecting critical aspects such as environmental protection, emergency preparedness, healthcare systems, water and sanitation infrastructure, and social safety nets in an attempt to save costs has proven to be a shortsighted strategy. The consequences of this neglect are now being felt on a massive scale.

Government of India “National One Health Mission”¹⁰

The creation of a National One Health Mission was approved on **July 20, 2022** by the Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC). Through this project, various ministries will work together to successfully fill any gaps while coordinating, advancing, and integrating all current One Health initiatives across the nation.



Mission

National One Health Mission, aims to coordinate across achieving overall **pandemic preparedness and integrated disease control** against priority diseases in both human and animal sectors, with early warning systems built on integrated surveillance system and response readiness for endemic as well as emerging epidemic or pandemic threat.

The mission will also address critical pillars of preparedness in the **form of targeted R&D** to develop important tools such as vaccines, diagnostics and therapeutics, readiness in terms of clinical care, streamline data and information linkages and access across sectors as well as close community participation to keep our readiness to respond.



Goals Of National One Health Mission: Joint Outbreak Investigation and Response Process



Implementing integrated disease surveillance within and across human, animal, and environmental sectors to address communicable diseases of zoonotic, transboundary animal diseases and diseases of epidemic/pandemic potential. This includes building early warning systems and integrating information coming from clinical and epidemiological data, information from various surveillance programs regardless of the agencies across sectors implementing it and deploying novel approaches such as environmental surveillance for timely and more accurate detection of diseases.



Environmental surveillance system- The One Health Mission envisions a robust network of sentinel-based surveillance that provides an ongoing real time picture of pathogens/disease spread that when linked to other sources of information, can be powerful adjunct to fill a major gap in our understanding of disease transmission dynamics across multiple diseases in human, animal husbandry and wildlife.

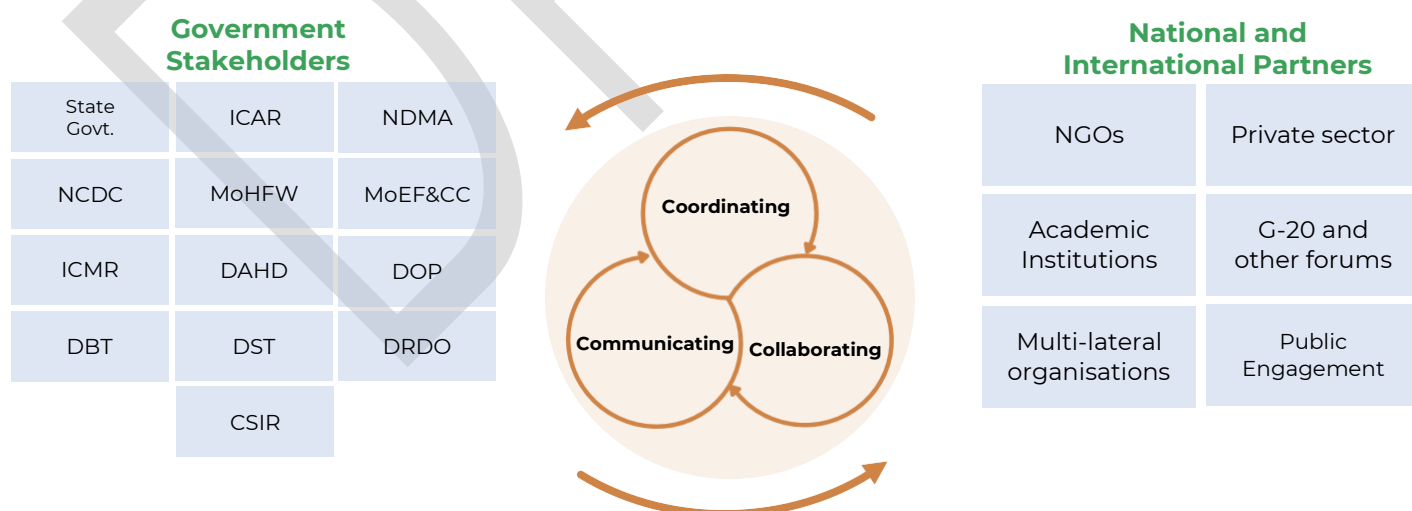


Develop robust outbreak investigation mechanisms for effective investigation and control of diseases across human, animal husbandry and wildlife sectors with framework for joint investigations. This includes building outbreak response teams at national levels linked to respective state programs and strengthening district level outbreak response teams such as Rapid Response Teams (RRTs), with clear framework for functioning as a connected system for routine disease control programs for addressing both endemic and emerging diseases.

Key Stakeholders For “National One Health Mission”¹⁰

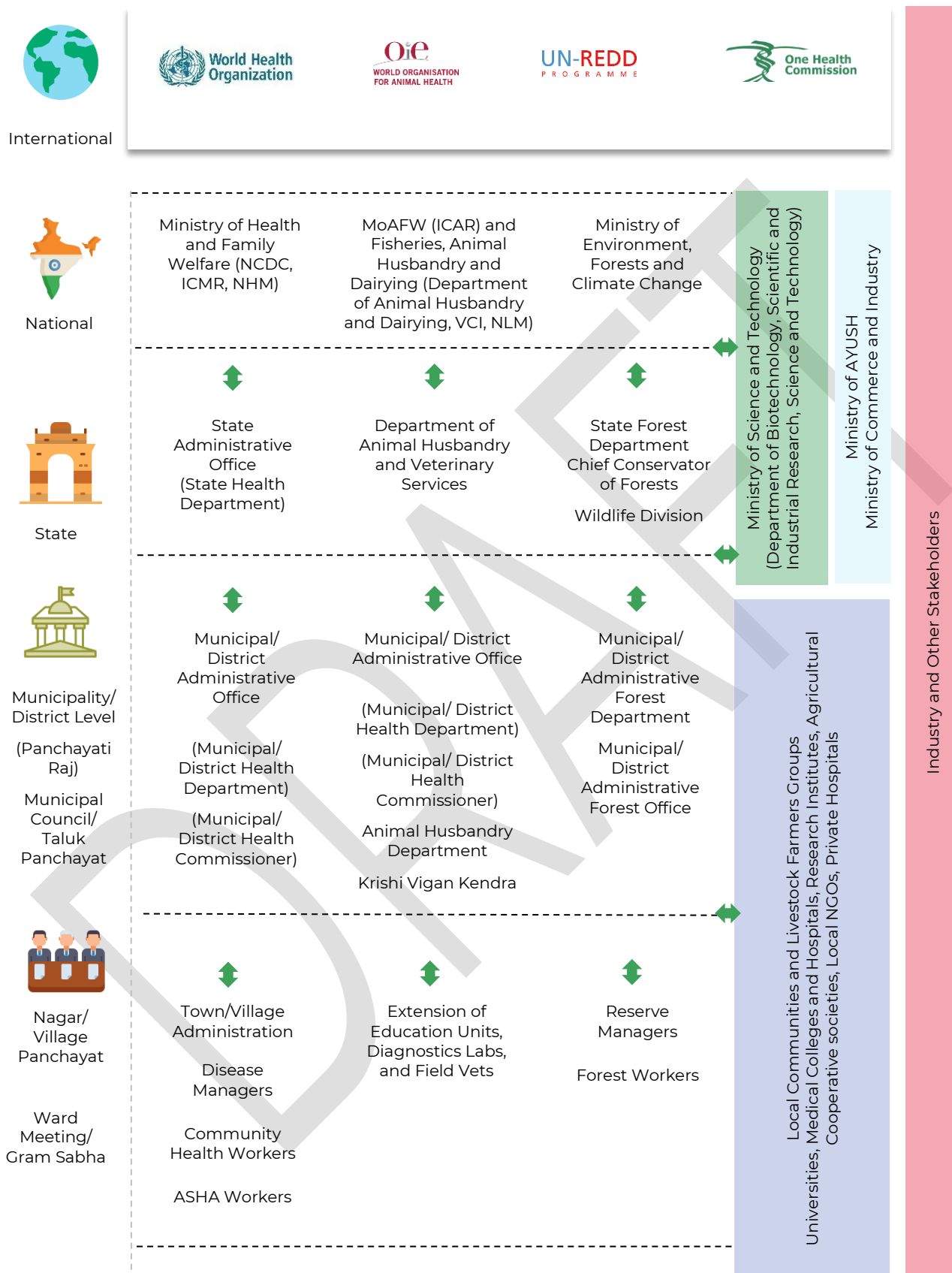
National One Health Mission is a collaborative effort, in which Government organisations and International Partners collaborate together. Collaboration efforts are aimed at the development of sustainable governance model to drive One Health approach.

Figure 2 – One Health Good Governance Model



ICAR; Indian Council of Agricultural Research, NDMA; National Disaster Management Authority, NCDC; National Centre for Disease Control, MoHFW; Ministry of Health & Family Welfare, MoEF&CC; Ministry of Environment, Forest & Climate Change, ICMR; Indian Council of Medical Research, DAHD; Department of Animal Husbandry & Dairying, DoP; Department of Pharmaceuticals, DBT; Department of Biotechnology, DST; Department of Science & Technology, DRDO; Defence Research & Development Organisation, CSIR; Council of Scientific & Industrial Research

Institutional Mechanism for One Health in India¹²



NCDC: National Centre for Disease Control, NHM: National Health Mission, MoAFW: Ministry of Agriculture and Farmers' Welfare, VCI: Veterinary Council of India, NLM: National Livestock Mission

¹² Asaaga et al. (2021). Operationalising the "One Health" approach in India: facilitators of and barriers to effective cross-sector convergence for zoonoses prevention and control.

India's One Health Platform¹¹

On September 22, 2021, the **Department of Animal Husbandry and Dairy** (DAHD) signed an MoU with the **Bill and Melinda Gates Foundation** to assist the Government of India in creating the **National One Health platform**. With the project's implementation in Uttarakhand and Karnataka, this ambitious project will be run for two and a half years. All the project's actions will result in the creation of a roadmap for scaling up the pilot initiatives and creating a One Health platform at the national level.

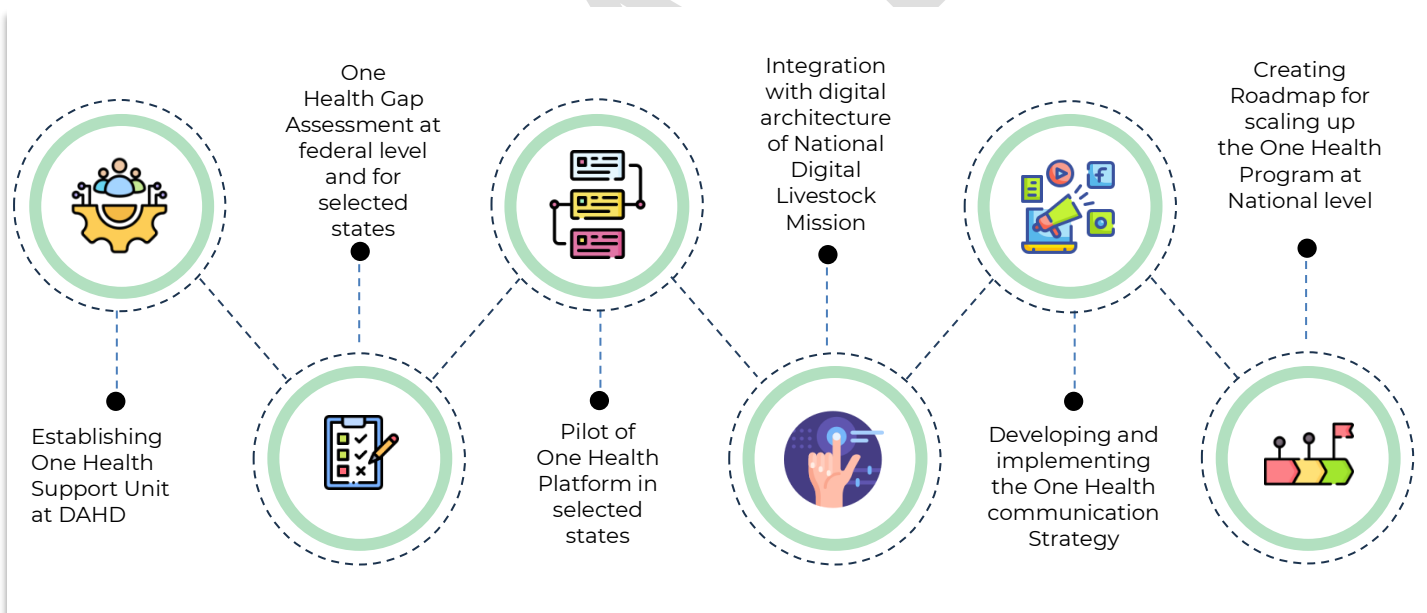


One Health Support Unit

To support development of "One Health Platform" the Department of Animal Husbandry and Dairy has established a **One Health Support Unit** made up of experts from a variety of fields, including veterinary medicine, animal husbandry, lab capacity evaluation, wildlife, epidemiology, animal husbandry economics, data standards/integration, administration and project management, and communications.

The Project Steering Committee is led by Secretary, Department of Animal Husbandry, Ministry of Fisheries, Animal Husbandry and Dairying, Government of India, and the Empowered Committee on Animal Health, led by Principal Scientific Advisor to Honourable Prime Minister of India, provide general direction to the team.

One Health Platform: Development Roadmap¹¹



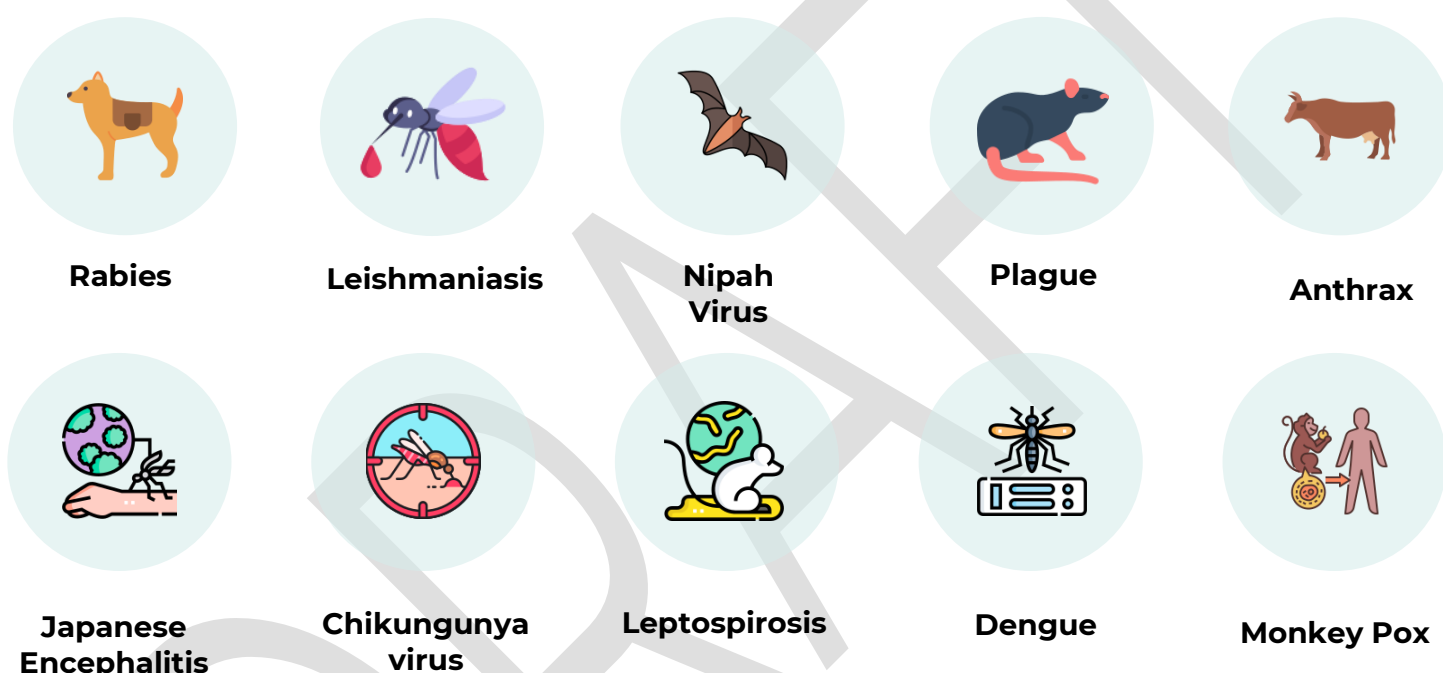
One Health Components

The One Health approach recognizes the interconnection and interdependence of human, animal, and environmental health, which are influenced by various factors. These factors include animal health affecting human health, environmental health affecting human health, antimicrobial resistance, and food safety.

1. Animal Health Affecting Human Health

It is an alarming fact that 2/3rd of the total infectious diseases affecting humans are transmitted from animals⁹

Zoonotic diseases are infectious diseases that can be transmitted between animals and humans. These diseases are caused by pathogens such as bacteria, viruses, fungi, and parasites. Wildlife often serves as the natural reservoir for these diseases, making them significant sources of public health emergencies. Most common **zoonotic diseases** affecting India:¹³



- Zoonotic diseases cause more than a billion infections and millions of deaths every year. 20,000 people deaths occur in India per year due to rabies which is a major public health concern.¹⁴
- 776 cases of Kala-azar (Leishmaniasis) were reported in 2022 in comparison to 1,166 cases in 2021, reporting a reduction of 33.44% of cases.¹⁵
- 54 districts (33 in Bihar, 11 in West Bengal, 4 in Jharkhand and 6 in Uttar Pradesh) are classified as endemic for Kala-azar.^{13,18}
- In recent history, a Nipah Virus outbreak was seen in Kozhikode, Kerala, in September 2021, where a 12-year-old boy presenting with acute encephalitis tested positive for Nipah Virus and died from the infection.¹⁶
- Plague is being monitored in south India trijunction, Maharashtra, Gujarat, H.P. and Uttarakhand. A localized outbreak of bubonic plague was reported from Uttarakhand in 2004 with 8 cases and 3 deaths.¹³
- In India the mosquito borne infection - Japanese encephalitis has been responsible for outbreaks in most of the states with a case fatality rate of 6-60%.¹⁷
- Chikungunya is endemic in 26 states and 6 UTs. During 2021, a total of 1,19,070 suspected cases of Chikungunya were reported.¹⁸
- Leptospirosis outbreak hit Mumbai and Thane districts in May 2000, and in August 2000 in Middle Kerala, resulting into great morbidity and significant mortality.¹³
- The first incidence of Monkeypox in India was reported on July 14, 2022, in Kerala, and following reporting of instances in Delhi and Kerala at intervals of 1-2 days. India has 10 confirmed cases (5 in Delhi and Kerala, and 1 each in Telangana, Bihar, and Uttar Pradesh), in addition to 8 suspected cases.¹⁹

¹³ Zoonosis Division National Centre for Disease Control. (2016). *Zoonotic Disease of Public Health Importance*.

¹⁴ One Health Support Unit. (n.d). Retrieved 20 September 2023.

¹⁵ National Centre for Vector Borne Diseases Control. (n.d) *Kala-azar Situation in India*. Retrieved 16 October 2023.

¹⁶ Yadav et al. (2022). Nipah Virus Outbreak in Kerala State, India Amidst of COVID-19 Pandemic.

¹⁷ Kulkarni R. (2018). Japanese Encephalitis: A Brief Review on Indian Perspectives.

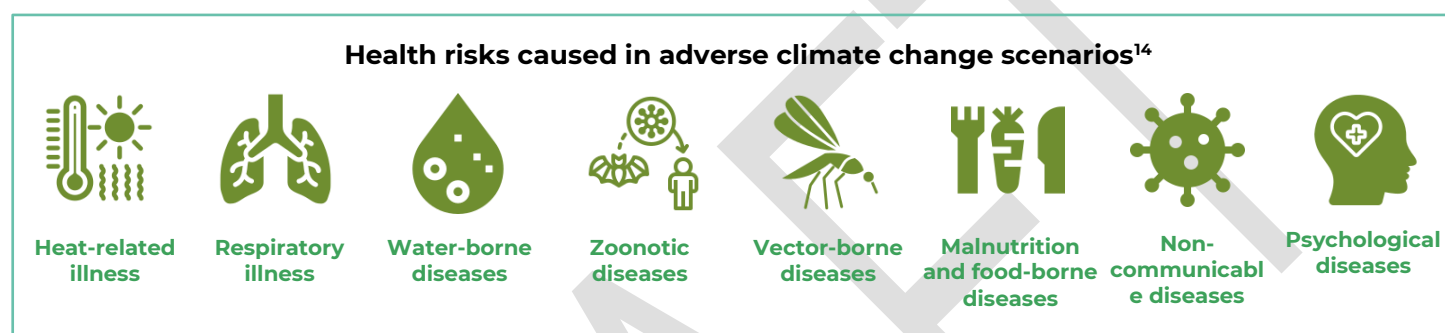
¹⁸ Ministry of Health and Family Welfare. (2023). *Annual Report*.

¹⁹ Singh T. (2022). Monkeypox: Current Situation in India: An Old Virus, A New Menace?

2. Environment Health Affecting Human Health

Climate change accelerates disease emergence, affecting human health systems and directly impacting animal health through extreme weather events. Rising temperatures impact disease vector distribution, contributing to the spread of vector-borne diseases. Habitat loss from climate change and human activities drives wildlife into new areas, increasing interactions with humans and livestock, and facilitating pathogen transmission. It is important to invest in disease surveillance for humans and animals while prioritising wildlife health management to safeguard ecosystems and wildlife.¹⁴

Climate change is already impacting in multiple ways on public health, manifesting as heightened deaths and illnesses stemming from more frequent extreme weather occurrences, like heatwaves, storms, and floods. It also disrupts food supply systems, contributes to the proliferation of zoonotic and disease vectors, and exacerbates mental health concerns. Moreover, climate change is eroding key social determinants of health, such as livelihoods, equity, access to healthcare, and support networks.¹⁴



- **Heat stress and air quality** are the two key areas considered to be vulnerable under the climate change scenarios.²⁰
- A total of 1,59,43,704 emergency OPD cases for Heat stroke were reported during 2022, which resulted in 534 deaths in 2022.²¹
- High disease burden and negative effects on a nation's economic growth are caused by **water-borne diseases**. Because of poor water supply management, particularly with regards to safe drinking water and sanitation, these diseases flare every year throughout the summer and rainy seasons.²²
- Approximately 37.7 million Indians are affected by waterborne diseases each year; 1.5 million children die from diarrhoea alone.²²
- These climate-related health risks disproportionately affect the most vulnerable and marginalized groups, encompassing women, children, ethnic minorities, impoverished communities, migrants, the elderly, and individuals with preexisting health conditions.
- India's population is at **risk for contracting malaria** except for those in the areas 1700 m above sea surface.
- More than 973 million persons are exposed to vector-borne malarial parasites in India. In 1998 an estimated 577,000 disability adjusted life years (DALYs) were lost due to malaria.¹⁹ 86,762 malaria cases and 23 deaths owing to malaria were reported in 2023.²³

²⁰ Ministry of Science and Technology. (2016). *Climate Change and Health*.

²¹ Ministry of Health and Family Welfare. (n.d.). *Impact of Extreme Heat on Health Risk in India*.

²² Kumar P. (2022). Prevalence and predictors of water-borne diseases among elderly people in India: evidence from Longitudinal Ageing Study in India, 2017–18.

²³ Dhara V. (2013). *Climate change & infectious diseases in India: Implications for health care providers*.

3. Antimicrobial Resistance (AMR): A Growing threat on Human Health

Multiple factors lead to the development of AMR in India. Rising incomes, uncontrolled access to antibiotics, including over-the-counter sales and sales without prescriptions, unfavourable financial incentives for healthcare professionals to prescribe antibiotics, have all contributed to the significant rise of AMR.²⁴

Antimicrobial Resistance Relationship

AMR can spread from person to person or between people and animals, and through environment



Antimicrobial
Resistance and
Humans



Antimicrobial
Resistance and
Agriculture and
Animal Farming



Antimicrobial
Resistance and
Environment

- Antimicrobial Resistance is a significant global health threat, with its impact particularly alarming in India. In 2010, India emerged as the **largest consumer of antibiotics for human health**, with consumption of 12,900 Crore units.²⁴
- The total defined daily dosage (DDD) usage of Antimicrobials has increased to 7,976 million in 2020 from 5,411 million in 2010 and DDD per capita usage of antimicrobials has risen to 5.74 in 2020 from 4.40 in 2010.²⁵
- India accounts for 3% of global consumption of antimicrobials in food animals and is fourth highest in the world, behind China (23%), the United States (13%) and Brazil (9%). The consumption of antimicrobials in the food and animals' sector in India is **further expected to double by 2030**.²⁴
- The burden of antimicrobial resistance AMR in Indian livestock and food animals remains inadequately documented, with limited comprehensive studies at the national level. The absence of significant regulations on antibiotic use for non-therapeutic purposes suggests that the emergence of AMR due to antibiotic overuse in the animal sector is likely an unmeasured issue in India.²⁴
- Contamination of environment with antibiotic residues occurring either through use in food animals or farms or through industrial effluents, is a rising concern. Issues of public sanitation, clean water, food animals being reared in unhygienic or unsanitary environments accumulate to more concern of AMR spreading.²⁴

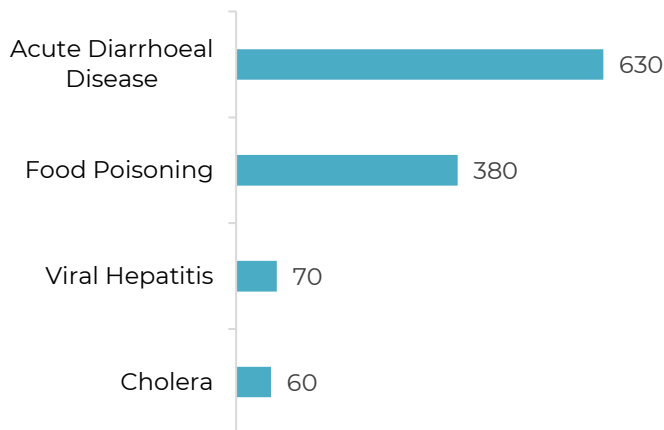
²⁴ World Health Organisation. (2016). *Antimicrobial Resistance and Containment in India*.

²⁵ The Centre for Disease Dynamics, Economics & Policy. (2021). *State of the World's Antibiotics*.

Food Borne Diseases (FBDs) and Infections

At least one of 12 people fall ill due to food-related infectious diseases in India. The number of FBDs is projected to grow by as much as 70% by 2030, largely driven by population growth and changing food consumption patterns.²⁶

Disease wise number of outbreaks reported under IDSP during 2020-2022²⁷



Total number of reported cases and deaths due to acute diarrhoeal diseases in India were 1,41,66,574 cases and 1,555 deaths in 2016 and 1,29,27,212 cases and 1,331 deaths in year 2017.²⁶

Human diarrhoeal diseases are mostly caused by bacterial food-borne infections (64%) followed by viral (24%) and protozoan (12%) infections.²⁶

Total of 70 districts reported outbreaks of Cholera during 2017-2021. During the same time period 84 outbreaks of Hepatitis were also reported.²⁸

²⁶ National Academy of Agricultural Sciences. (2020). *Food-borne Zoonotic Diseases*.

²⁷ IDSP. (2022). *Weekly Outbreaks Report*.

²⁸ IDSP. (n.d.). *Disease Wise Major Outbreak of the State*. Retrieved 31 October 2023.

Opportunities and Challenges

Opportunities



Technological Advancements⁶

India is moving fast towards adoption of software systems in healthcare. Similarly, India has immense potential for innovative, sustainable, and scalable healthcare technology to improve the lives of its citizens through one health approach.



Integrated One Health Surveillance Systems¹⁰

Implementation of disease surveillance across human, animal, and environmental domains to combat communicable diseases that encompass zoonoses, transboundary animal diseases, and diseases with epidemic or pandemic potential.



Strengthening Cross-sectoral Collaboration³⁰

Creation of Interoperable One Health Academic and in-service training programmes will strengthen the capacity of resource management and the collaboration between the experts of the three sectors environment, human and animal, for improved disease surveillance and addressing zoonotic threats, AMR, and food safety, while supporting One Health policies, legislations and interventions.



Establishing Data Sharing Mechanism³¹

By establishing data sharing mechanism between NCDC, DADF and ICAR (Through IHIP, NADRS, NADRES a comprehensive One Health approach can be implemented, leading to improved surveillance, early detection, and response to zoonotic diseases.



Establishing Joint Monitoring and Rapid Response Units³¹

Joint Risk Assessment and Joint Rapid Response teams at the National, State, and district levels are built to promptly mobilize resources, deploy diagnostic capabilities, and implement appropriate control measures in remote areas as well.



Strengthening of Laboratories⁹

Development of One Health Consortium by the Department of Biotechnology with the mission of building a network of laboratories to research about diseases and initiating collaborations between animal, environment and human health professionals. Strengthening of these labs in terms of capacity building, training workshops, research infrastructure will help in achieving the One Health goals.



Accepting Global Frameworks for One Health³⁰

Promoting One Health-based approach by following the Quadripartite's One Health Joint Plan of Action and contributing towards the One Health, One World strategic framework.

NCDC: National Centre for Disease Control, DADF: Department of Animal Husbandry and Dairying, ICAR: Indian Council of Agricultural Research, IHIP: Integrated Health Information Platform, NADRS: National Animal Disease Reporting System, NADRES: National Animal Disease Referral Expert System,

³⁰ World Health Organisation. (2022). *One Health Joint Action Plan*.

³¹ World Organisation for Animal Health. (2020). *One Health India*.

Challenges



Divergent Sectoral Priorities

Each sector involved in one health including animal, environment and human health operates with their own distinct priorities and objectives and can be influenced by factors such as funding, policies and other resources.



Operational Efficiency⁶

There is need of strong interdisciplinary network of sectors to improve surveillance and monitoring over humans, animals, and environment. Achieving seamless coordination can be challenging due to several factors. Each sector often has its own priorities, mandates, and bureaucratic structures, leading to potential conflicts in resource allocation and decision-making.



Limited Awareness⁶

There is a need to enhance the understanding of the One Health approach among everyone such as farmers, livestock managers, environmentalists, healthcare professionals, public health officials, policymakers, community leaders and advocacy groups, etc.



Limited Funding and Collaboration³²

The prime challenge in the implementation of various One Health activities is the availability of funds, and the difficulty in inter-ministerial collaboration towards a common funding strategy for One Health implementation.



Limited Inter-disciplinary Training³¹

Limited interdisciplinary training presents a notable challenge for the Government of India when it comes to implementing One Health. There is limited availability of scholarly resources or successful demonstration of one health approach by a nation with similar needs.



Human and Animal Health being State Subjects

As human and animal health are state subjects, it is handled by the state governments and each state may have their own regulations, policies and agendas for each department. Data for each sector is managed differently and it will be difficult to share the data in a uniform manner from all the states.



Rising AMR

Antimicrobial Resistance affects all the three components under one health. AMR can spread through food and the environment. Understanding the roles of food, agriculture, and the environment in AMR is vital to prevent infections and curb its further development and spread.

³² Dasgupta et al. (2021). Adopting an intersectoral One Health approach in India: Time for One Health Committees with The Indian journal of medical research.

Cross Cutting Themes



- **Significance:** Intersectoral Convergence aims to bring together different sectors and stakeholders to address the health, productivity, and conservation challenges in a holistic and integrated manner.
- **Current Efforts:** National One Health Mission, which is a cross-ministerial initiative to coordinate, support, and integrate all the existing One Health activities in the country and fill gaps where it is appropriate
- **Way Forward:** Strengthen the institutional mechanisms and capacities for effective collaboration and coordination among different sectors and levels of governance. Develop a common framework and indicators for monitoring and evaluating the outcomes and impacts of intersectoral convergence on One Health.



- **Significance:** Digital transformation is crucial for improving access, quality and affordability of health services, especially in the context of the COVID-19 pandemic and the emergence of zoonotic diseases.
- **Current Efforts:** The Government of India has launched several initiatives to promote digital health, such as the National Digital Health Mission, the Unified Health Interface, the e-Sanjeevani platform, and the Integrated Public Health Laboratories.
- **Way Forward:** Strengthening the data and information systems, enhancing the capacity and skills of the health workforce, ensuring accountability and transparency, and empowering the people to demand their rights to health.



- **Significance:** Health Equity addresses poor health outcomes, decreased access to and the quality of healthcare services, and greater vulnerability to infectious and chronic diseases.
- **Current Efforts:** Government has implemented various schemes and policies to enhance inclusivity in health, such as the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana, the National Health Mission, the Rashtriya Bal Swasthya Karyakram, etc.
- **Way Forward:** Identifying each stakeholder to build a framework with a multi-sectoral and collaborative approach that involves the participation of each community.



- **Significance:** Awareness is vital for advancing scientific knowledge, innovation and capacity building in the field of One Health, which aims to improve the health of humans, animals and the environment in an integrated manner.
- **Current Efforts:** The Government of India has established a National Expert Group on One Health. A National Institute of One Health at Nagpur, and a Centre for One Health Education, Research and Development at Gandhinagar are established to promote multi-sectoral, transdisciplinary collaboration and cooperation on One Health issues.
- **Way Forward:** To further sensitise the agenda on One Health in India, there is a need to enhance funding and infrastructure. Adopting best practices from national and global leaders



- **Significance:** Immunisation is a cross-cutting theme that aims to protect the health and well-being of humans and animals from preventable diseases.
- **Current Efforts:** Universal Immunization Programme (UIP), which is one of the largest public health programmes in the world, covering about 27 million new-borns and 30 million pregnant women annually. It provides free vaccines against 12 life-threatening diseases across the country.
- **Way Forward:** Improve the monitoring and evaluation of immunisation activities and outcomes, using data and evidence to inform decision-making and policy formulation.

Ministries and Departments

Inter ministerial and departmental collaboration is an important aspect of governance in India. Various ministries, departments, and organisations work together to achieve common goals and objectives.

 Ministries	 Department/ Organisation	 Function
 संसदीय कार्य मंत्रालय MINISTRY OF PARLIAMENTARY AFFAIRS Ministry of Parliamentary Affairs	Prime Minister's Science, Technology, and Innovation	Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC) approved the establishment of a National One Health Mission with a cross-ministerial effort to coordinate, support, and integrate all existing One Health activities in the country and fill gaps where appropriate
 स्वास्थ्य एवं संस्मरण मंत्रालय MINISTRY OF HEALTH AND FAMILY WELFARE Ministry of Health and Family Welfare	Department of Health and Family Welfare	The department is responsible for overall health policies and programs, including disease surveillance and control
	India Council of Medical Research (ICMR)	ICMR conducts medical research, including studies on zoonotic diseases and their impact on human health.
	National Centre for Disease Control (NCDC)	NCDC focuses on disease surveillance, outbreak investigations, and epidemiological research.
 Ministry of Environment, Forest and Climate Change	Indian Council of Forestry Research and Education (ICFRE)	The council conducts research related to forests and biodiversity which also includes the diseases affecting wildlife and forests.
 Ministry of Fisheries, Animal Husbandry and Dairying	Department of Animal Husbandry and Dairying	This department oversees various aspects of animal husbandry, including livestock health, welfare, and production. It plays a central role in disease surveillance, control, and management in domestic animals.



कृषि एवं किसान
कल्याण मंत्रालय
MINISTRY OF
AGRICULTURE AND
FARMERS WELFARE

Ministry of Agriculture and Farmers Welfare

Department of Agriculture and Farmers Welfare

Department is organised into 28 Divisions and has five attached offices and twenty-one subordinate offices which are spread across the country for coordination with state level agencies and implementation of Central Sector Schemes in their respective fields.

Indian Council of Agricultural Research (ICAR)

The Indian Council of Agricultural Research (ICAR) is the apex body for coordinating, guiding, and managing research and education in agriculture in the entire country.



विज्ञान एवं
प्रौद्योगिकी मंत्रालय
MINISTRY OF
SCIENCE AND
TECHNOLOGY

Ministry of Science and Technology

Department of Science and Technology

The department promotes scientific research and innovation in various fields, including environmental sciences, which are relevant to One Health concerns related to ecosystems and biodiversity.

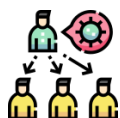
Department of Biotechnology

The department supports research and development in biotechnology, which includes projects related to health, agriculture, and the environment, contributing to a broader understanding of One Health issues.

Government Interventions

National One Health Mission¹⁰

On July 20, 2022, the Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC) approved the establishment of a National One Health Mission. This initiative involves a collaborative effort across multiple ministries to coordinate, support, and integrate all existing One Health endeavours in the country while addressing any existing gaps effectively. Following activities are prioritized under the mission²



Disease gap Prioritization Exercise

- A comprehensive collation of all the diseases of One Health priorities and available interventions across human, animal husbandry and wildlife sectors has been performed with extensive consultations.
- Gaps thus identified will be prioritized to serve as a guide for targeted R&D or epidemiological studies. This is envisioned to be a live database that will be continuously updated as the program evolves



Mapping of Existing Surveillance Programmes

- Prior to implementing integrating disease surveillance, a detailed mapping of all ongoing disease surveillance programs (both active and passive surveillance) across the sectors has been undertaken.
- Overlaps and potential opportunities for integrating them further will be taken up.



Mapping of Laboratory Capabilities

- For a better knowledge of the capabilities of the country, the laboratory capabilities of the current and planned BSL-2, 3 and 4 levels have been mapped.
- To enable efficient resource utilisation and capacity building for both routine disease control and managing new pandemics, this vital lab infrastructure will be better linked into a national network.



Mapping of Institutional Capabilities

- Mapping exercise was undertaken to assess the core capabilities of institutions of the stakeholder departments and highlight where they can contribute towards the Mission.
- This will further provide a collective understanding of the roles to be played in building a One Health ecosystem.

One Health Interventions By Other Ministries and Departments

Multiple ministries and departments in India are actively involved in advancing the progress and expediting the comprehension, adoption, and operationalization of the One Health approach across diverse domains. The heterogeneous expertise and resources possessed by these ministries will serve as catalysts for driving transformative change in the context of One Health initiatives.

National Programme on Climate Change and Human Health (NPCCHH)³³



It is an initiative which was launched to address the impact of climate change on human health through a holistic approach. The objectives of the programme include raising awareness about climate change effect on health among the general population, healthcare providers, and policymakers. It also focuses on building capacity for healthcare systems to mitigate climate-related illnesses and strengthening health preparedness and response through situational analysis.

NPCCHH aims to develop partnerships and synergy with other missions to ensure health considerations are integrated into the national climate change agenda. Additionally, it prioritises research capacity to fill knowledge gaps regarding climate's influence on human health. The overall strategy reflects India's commitment to safeguarding public health in the face of climate change by fostering awareness, resilience, and evidence-based action.

³³ National Centre for Disease Control. (2023). *National Programme on Climate Change & Human Health*. Retrieved 20 September 2023

Integrated Disease Surveillance Programme (IDSP)³⁴



IDSP was launched in 2004 with the aim to detect and respond to early warning signals of disease outbreaks through decentralised, IT-enabled systems, to enable the initiation of timely and effective public health actions. Training of rapid response team and setting up of laboratories and surveillance units in Centre, State/UTs and Districts.

Currently, there training of rapid response team and establishment of surveillance units have been completed for all the States/UTs. A portal has been created for data entry and analysis of reporting of disease outbreaks and to facilitate the units with other documents and training modules. About 90% of the districts in the country report weekly through the surveillance units about any disease outbreaks.

National Programme for Antimicrobial Resistance Containment³⁵



Controlled by National Centre for Disease Control (NCDC) the Programme was launched in 2013 to establish laboratory-based AMR Surveillance System, strengthen infection control practices and to generate awareness on AMR.

Under the Programme, National AMR Surveillance Network (NARS-Net) was developed to assess the magnitude and trends of AMR in various geographical regions of the country.

National AMR Surveillance Network (NARS-Net)

Under NARS-Net, the capacity building for network labs is fulfilled by including government medical colleges who are required to submit surveillance data on AMR



National Reference Labs (NRLs): There are two NRLs under this for bacterial pathogens and fungal pathogens.



Sentinel AMR Surveillance sites: Includes 40 labs in 31 states/UTs.

National Action Plan for Dog Mediated Rabies Elimination from India by 2030³⁶



It is a comprehensive initiative aimed at eradicating human rabies cases caused by dog bites. Its objectives include achieving zero human rabies deaths by 2030, implementing dog population management strategies such as mass vaccination and sterilization, and ensuring timely access to post-exposure prophylaxis (PEP) for dog bite victims. The plan emphasizes public awareness campaigns to educate communities about rabies prevention and responsible dog ownership.

It also strengthens disease surveillance systems, fosters inter-sectoral collaboration through a One Health approach, and trains healthcare providers in proper wound management and PEP administration. By integrating vaccination, education, and community engagement, India aims to eliminate the threat of rabies transmission from dogs and protect public health by the end of this decade.

Initially, Rabies control efforts were under National Rabies Control Programme which was in effect since the year 2008.

³⁴ Ministry of Health & Family Welfare. (2023). *Integrated Disease Surveillance Programme*. Retrieved 20 September 2023.

³⁵ National Centre for Disease Control. (2023). *National Programme for Antimicrobial Resistance Containment*. Retrieved 20 September 2023.

³⁶ National Centre for Disease Control. (2021). *National Action Plan for Dog Mediated Rabies Elimination from India by 2030*.

Inter-Sectoral coordination for Prevention and Control of Zoonotic diseases³⁷



The strategy involves enhancing inter-sectoral coordination by leveraging the existing surveillance system of the Integrated Disease Surveillance Programme (IDSP) to gather and organise data related to animal diseases for the establishment of early warning indicators. It also entails reinforcing the State Surveillance Units (SSU) under IDSP, fostering the development of trained personnel, raising awareness among professionals from various sectors, and conducting Information, Education, and Communication (IEC) campaigns to increase awareness among both the community and professionals. The objective is to establish ongoing collaboration that can facilitate outbreak investigations, response efforts, and the prevention and control of zoonotic diseases.

Programme for Prevention and Control of Leptospirosis³⁸



Leptospirosis is a public health concern in multiple Indian states, with frequent outbreaks mostly affecting young adult males. The Programme for Prevention and Control of Leptospirosis was launched with the strategy of strengthening of laboratories for early diagnosis, training manpower, strengthening of inter-sectoral coordination, creating awareness about the disease, and developing patient management facilities.

Launch of One Health Consortium⁹



Launched a Consortium for One Health at Department of Biotechnology-National Institute of Animal Biotechnology, Hyderabad where 27 participating centres, including major medical colleges (AIIMS Delhi, AIIMS Jodhpur, etc.) and veterinary universities (GADVASU, TANUVAS, etc.), are partnering to undertake surveillance of certain select zoonotic diseases.

CSIR- National Environmental Engineering Research Institute (CSIR-NEERI)³⁹



Initiative taken to prepare a strong and extensive cadre of human resources through various short-term and long-term academic, training, and skill development programmes to meet the environmental challenges of the 21st century such as air pollution, water pollution and waste management. By employing metagenomic techniques, which involve analysing the genetic material present in soil samples, the project aims to comprehensively assess the soil microbial communities and their diversity.

Setting UP BSL-IV Laboratory⁴⁰



Indian Council of Medical Research (ICMR) initiated the preparatory activities for the establishment of a dedicated infrastructure in the form of the National Institute of One Health (NIO) at Nagpur which will also house a BSL-IV laboratory.

One Health Support Unit⁴¹



A One Health Support Unit has been set up at the Department of Animal Husbandry and Dairying to oversee various one health pilot projects. The learnings from the pilot project will be used to develop a National One Health Framework.

³⁷ National Centre for Disease Control. (2023). *Inter-Sectoral coordination for Prevention and Control of Zoonotic Diseases*. Retrieved 20 September 2023.

³⁸ National Centre for Disease Control. (2015). *Programme for Prevention and Control of Leptospirosis*.

³⁹ Council of Scientific & Industrial Research. (2023). *National Environmental Engineering Research Institute*. Retrieved 20 September 2023.

⁴⁰ The World Bank. (2022). *Environmental and Social System Assessment*.

⁴¹ Department of Animal Husbandry and Dairying. (n.d). *One Health Support Unit*. Retrieved 20 September 2023.

Initiation of Soil Health Card Scheme⁴³



Ministry of Agriculture and Farmer's Welfare initiated Soil Health Card Scheme under which the soil health cards carrying crop-wise recommendations are issued to farmers. In this health card one parameter i.e., Soil Organic Carbon indirectly assesses the soil microbes required to sustain agricultural system which again represents the amalgamation of One Health principle.

Development of a National One Health Framework⁴⁴



Department of Animal Husbandry and Dairying (DAHD) under Ministry of Fisheries, Animal husbandry and Dairying launched pilot project in Karnataka and Uttarakhand, on 2022, for development of a national One Health Framework, aiming to collaborate with stakeholders from diverse sectors to enhance livestock health, human health, wildlife health, and environmental health through the utilization of technology and financial resources.

Animal Pandemic Preparedness Initiative and World Bank-funded Animal Health System Support for One Health (AHSSOH)⁴⁵



In April 2023, Union Minister for FAHD launched the Animal Pandemic Preparedness Initiative and World Bank-funded Animal Health System Support for One Health (AHSSOH) to enhance India's preparedness and response to potential animal pandemics, in line with the One Health approach.

⁴² National Portal of India. (2018). *Soil Health Card*. Retrieved 20 September 2023.

⁴³ Department of Animal Husbandry and Dairying. (2022). *Achievements*.

⁴⁴ The World Bank. (2023). *The Pandemic Fund*. Retrieved 20 September 2023.

Way Forward

India aims to strengthen and motivate health systems and stakeholders towards achieving the goal and vision of transitioning towards One Health, promoting a holistic approach of Health for all.



Raising Awareness and Fostering Ownership

One Health stakeholders will look to generate more awareness by driving campaigns and educational workshops. A Workshop on Environmental Surveillance for One Health, conducted on February 2023 at Indian Institute of Science, Bengaluru, aimed to build a shared knowledge of One Health Policies.⁴⁵



Joint Integrative Approach of Critical Pillars

Joint integrative projects, infrastructures, and work processes across the domains and themes of the One Health can be developed to strengthen the prevention, preparedness, detection and response capacities of emerging threats across human-veterinary-environmental sectors.



Identifying International Collaborations for One Health

There are several ongoing global best practices for successful implementation of disease programmes. To ensure sustainable implementation of One Health approach, India can incorporate more collaborations like Bill and Melinda Gates foundation with One Health Support Units, for developing National One Health Framework.¹¹



Monitoring Progress and Strengthening Data Systems

Enabling better coordination between different sectors and enhancing the monitoring of progress on a national scale and facilitating effective implementation at the local level. Appropriate indicators, parameters, and KPIs can be developed to strengthen data collection, monitor progress, measure outcomes, and ensure accountability.¹⁰



Developing Outbreak Investigation Mechanism

Formation of national-level outbreak response teams, connected to state programs, and strengthening district-level rapid response teams (RRTs). The integrated system will ensure the synchronized functioning of routine disease control initiatives, effectively addressing both persistent and emerging diseases.¹⁰



Integration of One Health with environment conservation strategies

Conserving natural environments supports biodiversity, curbing disease transmission and enhancing overall well-being. One Health Joint Plan of Action (OHJPA) states, to strengthen One Health, it's crucial to integrate environmental priorities by incorporating environmental data in decision-making, fostering mutual understanding among One Health stakeholders, and empowering environmental institutions for equal participation in decision-making processes.

One of the action plans under OHJAC is to develop ensure the integration of sessions aimed at in-service medical, public health and veterinary professionals, emphasising the significance of biodiversity conservation and interconnectedness between the health and environment, the role of environmental degradation in disease emergence and the vital aspect of merging the environmental sector and One Health partnerships.

⁴⁵ Office of PSA. (2023). *Workshop on Environmental Surveillance for One Health*.

Best Practices

1. Surveillance of Zoonoses and transboundary Pathogens, Assam⁴⁶

Background

Assam State is particularly susceptible to potential outbreaks in India due to several factors, including extreme climate events, unequal access to healthcare for rural communities, constant interaction among wildlife, livestock, and humans, and a rise in migration. Despite Assam's susceptibility to pandemics, there is a limited number of solution-based approaches and collaborations in place for emergencies and outbreaks, particularly in Northeastern India and Assam State.

One Health Approach Applied

- In October 2021, the Department of Biotechnology supported a Program on 'One Health' involving 27 organisations, including Assam Agricultural University and wildlife agencies. This program aims to conduct surveillance of zoonoses and transboundary pathogens in India, including the northeastern region.
- Furthermore, Assam Agricultural University, along with its two constituent Veterinary colleges in Khanapara, Guwahati, and Lakhimpur with Departments of Veterinary Microbiology and Veterinary Public Health, offers research and diagnostic support to the State Government. Additionally, they provide training to veterinary staff.
- The Food Safety and Standards Authority of Assam (FSSAA) began enforcing hygienic and sanitary practises in response to the reported instances of African swine fever in Assam.

Figure 3 – Snippet of Assam during flood



⁴⁶ World Bank Group. (2022). *One Health- Assam case study*.

2. Bengaluru: Swasti “Precision Health Platform”⁴⁷

Background

In March 2021, Swasti's ground breaking initiative, the Precision Health platform, was launched as a COVID-19 wastewater surveillance project. The platform's primary focus was to track and monitor the presence of the virus in wastewater as an early indicator of potential outbreaks and to support public health responses.

One Health Approach Applied

- Among the significant initiatives hosted under this platform, the Bengaluru City-wide COVID-19 Surveillance program stands out. This program has become instrumental in monitoring the spread of COVID-19 within the city allowing officials to make well-informed decisions regarding pandemic preparedness and response strategies.
- Additionally, the platform also spearheads the Genetic Variants Surveillance initiative, which tracks and analyses the genetic variations of the virus. This critical undertaking aids in understanding viral mutations and assists in refining public health measures accordingly. Through the Precision Health platform, city officials and health authorities gain valuable insights and data essential for their decision-making processes related to COVID-19. Serving as an early warning system, the platform plays a pivotal role in detecting potential outbreaks swiftly and supporting effective pandemic management.

Figure 4 – Wastewater Surveillance



⁴⁷ Swasti. (n.d). *One Health*. Retrieved 20 September 2023.

3. Response to Nipah Virus, Kerala⁴⁸

Background

Nipah virus infection is a newly emerging zoonotic disease that affects both animals and humans, with fruit bats being its natural host. The infection can result in severe respiratory distress, encephalitis, and seizures, and in some cases, it may rapidly progress to a coma within 24 to 48 hours.

In response to the emergence of Nipah Virus Disease cases in Kerala during 2018 and 2019, a multi-disciplinary Central team was dispatched by the Union Health Minister. The team, comprising public health experts, microbiologists, neurologists, and other infectious disease specialists, aided local state authorities in investigating and managing both outbreaks.

One Health Approach Applied

- To investigate the outbreaks in 2018 and 2019, the District Rapid Response Teams collaborated with the central multi-disciplinary team. The teams conducted active case searching and contact tracing through house-to-house visits. The state's Nipah cell was established for efficient contact tracing and ensuring home quarantine. Control rooms were set up at the State and District Headquarters, with daily review meetings conducted by the Core Committee, chaired by the State Health Minister and District Collectors.
- An action plan was developed to train healthcare staff, including a simulation exercise. Risk communication messages were delivered to the community, public, stakeholders, and partners. Training sessions were organised for infection control practices, specimen transportation, dead body handling, and other essential aspects.
- The Kerala outbreak highlighted the reality of new and emerging diseases and underscored the need for preparedness and robust health systems to effectively respond to such health challenges.

Figure 5 – District Rapid Response Teams



⁴⁸ Singhai M et al.(2021). Nipah Virus Disease: Recent Perspective and One Health Approach with Annals of Global Health.

Recommended Readings

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